

CoAlexist Studio - Recent Enhancements



Major UI Improvements (December 30, 2025)

This document outlines the latest enhancements to CoAlexist Studio, making it more visual, intuitive, and powerful for non-technical users.



What's New

1. 🖌️ Removed Scanlines

What changed:

- Removed the default scanline overlay effect from both dark and light modes
- Cleaner, less distracting interface

Why:

- User feedback indicated the scanlines were visually distracting
- Provides a cleaner workspace for design work

Location:

- `index.html` - Lines 101-111 (dark mode) and 50-53 (light mode)
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2. 📏 Enhanced Numeric Inputs

What changed:

- Replaced plain text inputs for numbers with a visual control system:
- **Sliders** for smooth value adjustment
- **Stepper buttons** (▲/▼) for precise incremental changes
- **Unit dropdowns** (px, %, em, rem, vw, vh)
- Smart range detection based on property type

Features:

- Sliders automatically adjust range based on unit:
- `px` : 0-1000
- `%` : 0-200
- `em / rem` : 0-10 (step 0.1)
- `opacity` : 0-1 (step 0.01)
- Real-time synchronization between slider, input, and canvas
- Visual feedback on hover and click

Why:

- Makes the editor more accessible for users who are “bad with numbers”
- Visual controls are more intuitive than typing values
- Faster workflow with sliders for rough adjustments and steppers for fine-tuning

Location:

- `main-app.js` - Lines 1702-1770 (makeNumInput function)
- `index.html` - Lines 763-795 (styling)

Applies to:

- Width, Height, Top, Left, Right, Bottom
 - Margin, Padding, Min-Width, Max-Width
 - Font Size, Line Height, Letter Spacing
 - Border Width, Border Radius
 - Opacity, Z-Index
 - All other numeric CSS properties
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3. 🌐 Neocities Pages Panel

What it does:

- Lists all HTML files from your Neocities site (coaexist.wtf)
- Load any page directly into the editor for editing
- Open pages in new tab for preview
- Refresh button to update file list

Features:

- Real-time file listing via Neocities API
- Visual file browser with hover effects
- Quick actions: Edit (✏️) and Open (🔗)
- Error handling with friendly messages

How to use:

1. Open the sidebar (right panel)
2. Find the “🌐 Neocities Pages” panel
3. Click on any HTML file’s ✏️ button to load it into the editor
4. Click 🔗 to open the live page in a new tab
5. Use ↻ Refresh to reload the file list

API Integration:

- Uses Neocities API key: `95cba50ce217a25db2e85800e178044e`
- Fetches files from: `https://neocities.org/api/list`
- Opens pages from: `https://coaexist.neocities.org/`

Location:

- `src/neocities-pages-panel.js` - Main panel logic
 - Automatically loads after 300ms on page load
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4. 🎨 Theme Extractor Panel

What it does:

- Analyzes any webpage (via URL or current canvas) and extracts:
- Color palette (up to 8 colors)
- Font families used
- Common design patterns
- Shows extracted theme as a visual preview

- Apply theme to your current canvas
- Save theme as a Style Chip preset

Features:

- **Analyze URL:** Extract theme from any website
- **Analyze Current Canvas:** Extract theme from your current design
- **Visual Color Swatches:** Click to copy hex codes
- **Font Preview:** See fonts with sample text
- **Apply Theme:** One-click theme application
- **Save as Preset:** Add extracted theme to Style Chips

How to use:

1. Open the sidebar (right panel)
2. Find the “🎨 Theme Extractor” panel
3. **Option A:** Enter a URL and click “🔍 Analyze URL”
4. **Option B:** Click “🎯 Analyze Current Canvas”
5. View the extracted colors and fonts
6. Click “✨ Apply Theme” to use it on your canvas
7. Click “💾 Save Preset” to add it as a Style Chip

Technical Details:

- Parses HTML and CSS to extract colors
- Supports hex, rgb, and rgba color formats
- Extracts font-family declarations from stylesheets and inline styles
- Uses CORS proxy (allorigins.win) for cross-origin requests
- Normalizes colors to hex format

Location:

- `src/theme-extractor-panel.js` - Main panel logic
- Automatically loads after 400ms on page load

5. ✅ Verified Style Chips Panel

Status:

- Panel is visible and functional
- 5 preset styles available:
 1. 💎 **Glassy** - Glass morphism effect
 2. 📼 **VHS Burn** - Retro VHS aesthetic
 3. 💻 **Terminal** - Command line style
 4. 🌊 **Frutiger Aero** - Windows Vista era
 5. ✨ **McBling** - Y2K maximalist style

Features:

- Hover to preview style
- Click to apply to selected element
- Visual feedback on application
- Separate from CSS editor (as requested)

Location:

- `src/style-chips.js` - Panel logic
- Loads after 200ms on page load

File Structure

```
/home/ubuntu/coaixist-studio-package/  
├── index.html  
├── # Main HTML (scanlines removed, stepper styles added)  
├── main-app.js # Enhanced makeNumInput function  
├── src/  
│   ├── style-chips.js # Style presets panel  
│   ├── neocities-pages-panel.js # NEW: Neocities integration  
│   ├── theme-extractor-panel.js # NEW: Theme analysis tool  
│   ├── world-graph-panel.js # World graph visualization  
│   ├── mood-status-bar.js # Status bar enhancements  
│   └── wysiwyg-resize-handles.js # Visual resize handles  
├── package.json # Dependencies  
├── vite.config.ts # Vite configuration  
└── tsconfig.json # TypeScript configuration
```

Getting Started

Running the Studio

```
cd /home/ubuntu/coaixist-studio-package  
  
# Install dependencies  
npm install  
  
# Start development server  
npm run dev  
  
# Server runs on http://localhost:3000
```

Quick Test (Python Server)

```
cd /home/ubuntu/coaixist-studio-package  
python3 -m http.server 8080  
  
# Open http://localhost:8080 in your browser
```

User-Friendly Design Principles

All enhancements follow these principles:

1. **Visual over Textual:** Sliders and steppers instead of plain inputs
2. **Immediate Feedback:** Real-time updates as you adjust values
3. **Progressive Disclosure:** Advanced features hidden until needed
4. **Error Tolerance:** Graceful error handling with friendly messages
5. **Consistency:** Same interaction patterns across all panels

Future Enhancements

Potential areas for future development:

- [] Save to Neocities directly from editor
 - [] More style presets in Style Chips
 - [] Theme variations (light/dark mode generation)
 - [] Component library from extracted patterns
 - [] AI-powered theme suggestions
 - [] Collaborative editing features
 - [] Version history and undo/redo improvements
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Developer Notes

Adding New Panels

To add a new panel to the sidebar:

1. Create a new file in `src/` (e.g., `my-panel.js`)

2. Follow the pattern:

```
```javascript
(function() {
 'use strict';

 function createMyPanel() {
 const sidebar = document.querySelector('.sidebar-content');
 // ... panel creation logic
 }

 // Initialize
 if (document.readyState === 'loading') {
 document.addEventListener('DOMContentLoaded', () => setTimeout(createMyPanel, 500));
 } else {
 setTimeout(createMyPanel, 500);
 }
})();
```

3. Add script tag to `index.html`: `html`

...

### Styling Conventions

- Use CSS variables: `var(--magenta)` , `var(--cyan)` , `var(--text-main)`
  - Follow BEM-like naming: `.panel-name__element--modifier`
  - Keep z-index values consistent
  - Support both light and dark modes
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## Known Issues

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- None at this time!
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## Performance Notes

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- All panels use deferred loading (200-500ms delays)
  - API calls are cached when possible
  - CORS proxy adds latency for external URL analysis
  - Large color palettes are limited to 8 colors max
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## Credits

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- **Original CoAlexist Studio:** Amazing foundation
  - **Enhancements:** Implemented based on user feedback
  - **APIs:** Neocities API for file management
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## License

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Same as CoAlexist Studio

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## Support

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For issues or feature requests, check the main CoAlexist Studio repository.

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**Last Updated:** December 30, 2025

**Version:** 1.1.0

**Commit:** 4401507